



MOHAMED MAAMOUN MAROUANI

Engineering Student • Industrial Electronics

Embedded Systems • Instrumentation • Automation •
Energy

PFE ENGINEERING PORTFOLIO

Selected projects, practical industrial exposure and technical capabilities.

ENISo
École Nationale d'Ingénieurs de
Sousse

FOCUS
Mobility & Renewable Energy

OBJECTIVE
4-6 month PFE internship

“Building practical engineering solutions at the intersection of hardware, software, measurement and control.”

[VIEW LINKEDIN PROFILE](#)

01 | ABOUT ME

I am an Industrial Electronics engineering student at ENISO, with a technical orientation toward embedded systems, instrumentation, industrial automation and energy-related applications.

My projects connect physical systems to digital intelligence: acquiring measurements, communicating with sensors and controllers, processing data, implementing monitoring or control functions, and validating the complete system.

I am looking for a PFE environment where I can contribute to technical projects while strengthening my experience in real industrial systems, testing, integration and engineering problem solving.

CORE ENGINEERING PROFILE

Embedded Systems	STM32 • ESP32 • Arduino • C/C++ • Python
Communication	CAN • Modbus RTU • UART • I ² C
Instrumentation	Voltage/current measurement • sensors • alarms • real-time acquisition
Automation	PLC concepts • ladder logic • HMI • process sequencing
Tools	MATLAB/Simulink • Proteus • STM32CubeIDE • Arduino IDE • VS Code

ENGINEERING INTERESTS

Instrumentation & sensing • Embedded control • Industrial automation • Data acquisition • Testing & validation • Energy systems • Mobility technologies • Clean-energy applications

02 | TECHNICAL TOOLKIT

Embedded	STM32 • ESP32 • Arduino • C/C++	Firmware, peripherals, acquisition, control
Protocols	CAN • Modbus RTU • UART • I ² C	Industrial and automotive communication
Automation	PLC • Ladder Logic • HMI	Process sequencing and machine control
Measurement	Voltage • Current • Power • Energy • Sensors	Monitoring and instrumentation
Software	Python • MATLAB/Simulink • Proteus	Analysis, modeling, simulation, prototyping
Interfaces	Web dashboards • LCD/HMI	Real-time visualization and operator feedback

ENGINEERING APPROACH

Physical process → sensing → communication → embedded logic → visualization → testing → validation.

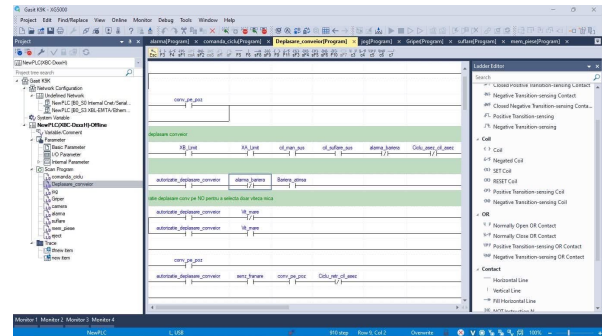
I aim to understand the complete system rather than only one layer of it, while remaining comfortable learning new tools and technologies.

03 | INDUSTRIAL INTERNSHIP EXPERIENCE

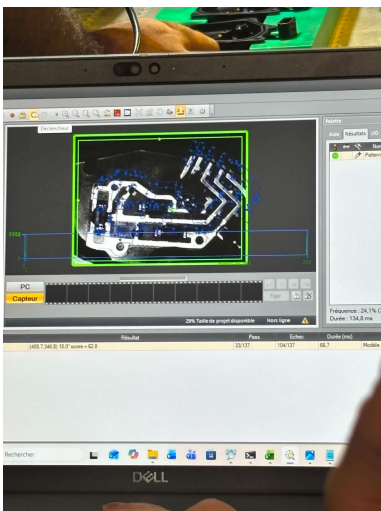
Selected visual documentation from my industrial internship, showing exposure to automation equipment, conveyor/process systems, PLC programming and machine-vision inspection.



Industrial conveyor / process equipment
Control cabinet, HMI, indicators and conveyor-based handling.



PLC programming environment
Ladder-logic programming and process sequencing.



Machine-vision inspection
Camera-based inspection interface and result analysis.



Industrial component
Physical hardware observed in an industrial inspection context.

This experience strengthened my interest in practical automation, industrial equipment, process control and inspection systems.

04 | FEATURED PROJECT — AUTOMOTIVE CAN DIAGNOSTICS

Intelligent Automotive Diagnostic System — CAN Bus Communication, Fault Analysis & Web Monitoring

ENISo Academic Project • 2025-2026

A team project focused on acquiring and analyzing automotive CAN data with an ESP32 + MCP2515 architecture, then presenting information through a real-time web dashboard.

Hardware	ESP32 • MCP2515 • CAN/OBD interface
Communication	CAN Bus • Wi-Fi
Dashboard	HTML/CSS/JavaScript • real-time monitoring
Diagnostics	CAN frame acquisition • vehicle parameters • DTC handling
AI extension	Local LLM experimentation for interpreting diagnostic codes and presenting troubleshooting guidance

ENGINEERING VALUE

This project combines automotive communication, embedded programming, networking, data visualization and diagnostic logic in one integrated system.

PROJECT POST • CAN

PROJECT POST • CAN 2

The LinkedIn posts contain the original project media and demonstration material.

05 | FEATURED PROJECT — STM32 ENERGY MONITORING

Industrial Electrical Energy Monitoring — STM32, Modbus RTU & HMI

ENISo Academic Project

An embedded monitoring platform designed to acquire electrical quantities and transform measurements into real-time power and energy information.

Controller	STM32F407VG
Measurement	Voltage • current • power • energy
Communication	UART • Modbus RTU • I ² C
Interface	LCD/HMI • status indicators • alarms
Software	Embedded C • STM32CubeIDE • FreeRTOS

ENGINEERING VALUE

The project developed practical understanding of measurement chains, embedded processing, industrial communication, task management and operator feedback.

[PROJECT POST • ENERGY](#)

[PROJECT POST • STM32](#)

06 | ADDITIONAL ENGINEERING PROJECTS

ESP32 IOT MONITORING

Connected gas and noise monitoring system combining ESP32 sensor acquisition, Wi-Fi communication, threshold detection and real-time web visualization.

Platform	ESP32 • sensors • Wi-Fi
Functions	Real-time acquisition • thresholds • alerts • dashboard
Focus	Connected sensing and low-latency monitoring

MOBILE ROBOTICS PLATFORM

ENISo team project involving an all-terrain four-wheel platform designed to support motors, controllers, sensors and battery systems, with a focus on platform development and future system integration.

OBSTACLE-AVOIDANCE ROBOT

Arduino-based mobile robot using ultrasonic sensing for obstacle detection and autonomous avoidance.

MATLAB / SIMULINK — MODELING & CONTROL

Academic work involving system modeling, simulation, electrical quantities, power calculations and control-oriented analysis.

PROJECT MINDSET

Across these projects, I have worked on the path from physical hardware and sensors to embedded logic, communication, monitoring and system-level validation.

07 | PFE OBJECTIVE & CONTACT

I am seeking a 4-6 month final-year engineering internship where I can contribute to a technical team and gain deeper practical experience in complex engineering systems.

AREAS OF INTEREST

- Embedded systems and real-time control
- Instrumentation and sensor integration
- Industrial automation and PLC/HMI environments
- Data acquisition and monitoring
- Testing, validation and system integration
- Energy systems and clean-energy technologies
- Electromechanical and mobility-related systems

MY APPROACH

I am motivated by projects where engineering decisions connect hardware constraints, software behavior, measurements, communication and real operating conditions. I enjoy learning from experienced engineers and turning technical requirements into working prototypes and validated systems.

MOHAMED MAAMOUN MAROUANI

Engineering Student — Industrial Electronics

ENISo • Sousse, Tunisia

PFE availability • 4-6 months

CONNECT ON LINKEDIN